E13e - Time-resolved spectroscopy and impedances

Group 22: Laiba Rehan Zahid - 3771917 (50%), Kaiting Chang - 3767255 (50%)

June 24, 2026

1 Introduction

The aim of this experiment was to investigate the time-dependent discharge behaviour of capacitors in RC and RLC circuits and to use these measurements to determine the frequency-dependent impedance and phase shift of the circuits. Instead of applying sinusoidal voltages at many individual frequencies, the circuits were excited by a switching process and the resulting time-domain response was measured using a Picoscope. The frequency-domain behaviour was then obtained from the recorded signals using Fourier-transform methods.

In the first part of the experiment, the discharge of a capacitor in an RC circuit was measured for five different resistances. The capacitor voltage was recorded as a function of time and fitted with an exponential decay function. From this, the characteristic time constant of the circuit was determined and compared with the theoretical value. Since the capacitor was charged through a fixed charging resistor and discharged through the selected resistance, the effective resistance during discharge was taken into account in the analysis. The measured time-domain response was also used to determine the impedance magnitude and phase shift of the RC circuit, which were then compared with the theoretical expressions for a series RC circuit.

In the second part of the experiment, the capacitor discharge was measured in an RLC series circuit for five different resistances. In this case, the discharge produced a damped oscillation, caused by the periodic exchange of energy between the capacitor and the inductor. The decay of the oscillation is due to energy dissipation in the resistive parts of the circuit. By fitting the measured oscillations, the damping coefficient and oscillation frequency were determined. Since the inductance of the coil was not known directly, it was estimated from the measured oscillation frequencies. The resulting impedance magnitude and phase shift of the RLC circuit were then calculated and compared with the theoretical behaviour of a damped series resonant circuit.

Overall, this experiment demonstrates how time-resolved measurements can be used to characterize AC circuits. The measured discharge curves provide information not only about the transient behaviour of the circuits, but also about their frequency-dependent impedance and phase response. Possible deviations from the ideal theoretical models are expected due to component tolerances, the internal resistance of the coil, the finite resolution of the Picoscope, and the non-ideal behaviour of the switching circuit.

2 Theoretical Analysis

2.1 Overview

The experiment is based on the relation between the time-domain response of an electrical circuit and its frequency-dependent impedance. When a capacitor is discharged through an RC or RLC circuit, the measured voltage signal contains information about the characteristic time constants, damping, resonance behaviour, impedance magnitude, and phase shift. In this section, the theoretical expressions used for the analysis of the RC and RLC circuits are derived.

For a harmonic signal, the voltage and current can be written as complex quantities proportional to

$$e^{i\omega t},$$

where

$$\omega = 2\pi f$$

is the angular frequency. The complex impedance is defined as

$$Z = \frac{U}{I} = |Z|e^{i\theta},$$

where $|Z|$ is the impedance magnitude and θ is the phase shift between voltage and current.

2.2 Impedance of Basic Circuit Elements

For an ohmic resistor, Ohm's law gives

$$U = RI,$$

so the impedance of a resistor is

$$Z_R = R.$$

For an inductor, the voltage is proportional to the time derivative of the current:

$$U = L\frac{dI}{dt}.$$

Using $I(t) = I_0 e^{i\omega t}$, one obtains

$$\frac{dI}{dt} = i\omega I,$$

and therefore

$$U = i\omega LI.$$

Thus, the impedance of the inductor is

$$Z_L = i\omega L.$$

For a capacitor,

$$Q = CU$$

and

$$I = \frac{dQ}{dt}.$$

Since $Q = CU$, this gives

$$I = C\frac{dU}{dt}.$$

For $U(t) = U_0 e^{i\omega t}$,

$$I = i\omega CU.$$

Therefore,

$$Z_C = \frac{U}{I} = \frac{1}{i\omega C}.$$

2.3 RC Circuit

For a series RC circuit, the total impedance is the sum of the resistor and capacitor impedances:

$$Z = Z_R + Z_C.$$

Using the expressions above,

$$Z = R + \frac{1}{i\omega C}.$$

Since $1/i = -i$, this becomes

$$Z = R - \frac{i}{\omega C}.$$

The impedance magnitude is therefore

$$|Z| = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}.$$

The phase shift is the argument of the complex impedance:

$$\theta = \arctan\left(\frac{\mathrm{Im}(Z)}{\mathrm{Re}(Z)}\right).$$

Thus, for the RC circuit,

$$\theta = -\arctan\left(\frac{1}{\omega RC}\right).$$

At low frequencies, the term $1/(\omega C)$ is large, so the capacitor strongly increases the impedance and the phase shift approaches $-90°$. At high frequencies, the capacitive impedance becomes small and the total impedance approaches R, while the phase shift approaches $0°$.

2.4 RC Discharge

For a capacitor discharging through a resistance, the capacitor voltage obeys

$$U_C(t) = U_0 e^{-t/\tau},$$

where the time constant is

$$\tau = RC.$$

In this experiment, the capacitor was charged through a fixed charging resistor $R_{\text{ch}} = 22\,\text{k}\Omega$. During discharge, the effective resistance seen by the capacitor is therefore the parallel combination of the discharge resistor R and the charging resistor:

$$R_{\text{eff}} = R \parallel R_{\text{ch}} = \frac{RR_{\text{ch}}}{R + R_{\text{ch}}}.$$

Therefore, the theoretical time constant used for comparison is

$$\tau_{\text{theory}} = R_{\text{eff}} C.$$

Since the capacitor may not discharge exactly to zero because of the charging path, the measured voltage is fitted with

$$U_C(t) = U_\infty + Ae^{-t/\tau},$$

where U_∞ is the final voltage offset and A is the initial amplitude of the exponential decay.

The Fourier transform of the normalized exponential discharge function

$$f_D(t) = e^{-t/\tau}$$

is

$$\hat{f}_D(\omega) = \frac{\tau}{1 + i\omega\tau}.$$

For the RC series circuit, the admittance relation used in the analysis is

$$\frac{R}{Z_s} = i\omega \hat{f}_D(\omega).$$

This relation allows the impedance magnitude and phase shift to be determined from the measured time-domain signal.

2.5 RLC Series Circuit

For the RLC series circuit, Kirchhoff's loop rule gives

$$U_L + U_R + U_C = V(t).$$

Using

$$U_L = L\frac{dI}{dt}, \qquad U_R = RI, \qquad U_C = \frac{Q}{C},$$

and $I = dQ/dt$, the differential equation becomes

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = V(t).$$

After the supply voltage is switched off, $V(t) = 0$, so

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0.$$

Dividing by L gives

$$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{1}{LC}Q = 0.$$

The natural angular frequency and damping coefficient are defined as

$$\omega_0 = \frac{1}{\sqrt{LC}},$$

and

$$\delta = \frac{R}{2L}.$$

The differential equation can therefore be written as

$$\frac{d^2Q}{dt^2} + 2\delta\frac{dQ}{dt} + \omega_0^2 Q = 0.$$

For the underdamped case, where $\delta < \omega_0$, the solution is a damped oscillation:

$$Q(t) = Q_0 e^{-\delta t}\cos(\omega_d t + \phi),$$

where the damped angular frequency is

$$\omega_d = \sqrt{\omega_0^2 - \delta^2}.$$

Since the measured capacitor voltage is proportional to charge,

$$U_C(t) = \frac{Q(t)}{C},$$

the voltage signal can also be described by a damped oscillation:

$$U_C(t) = U_{\text{off}} + Ae^{-\delta t}\sin(\omega_d t + \phi).$$

This fit function is used to determine δ and ω_d from the measured RLC discharge curves.

The unknown inductance can then be calculated using

$$\omega_0^2 = \omega_d^2 + \delta^2$$

and

$$L = \frac{1}{C\omega_0^2}.$$

2.6 RLC Impedance and Phase Shift

The impedance of a series RLC circuit is

$$Z = Z_R + Z_L + Z_C.$$

Substituting the individual impedances gives

$$Z = R + i\omega L + \frac{1}{i\omega C}.$$

Therefore,

$$Z = R + i\left(\omega L - \frac{1}{\omega C}\right).$$

The impedance magnitude is

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}.$$

The phase shift is

$$\theta = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right).$$

At resonance,

$$\omega L = \frac{1}{\omega C},$$

so the inductive and capacitive reactances cancel. The resonance angular frequency is therefore

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

At this frequency, the impedance is minimal and approximately equal to the total resistance:

$$Z \approx R.$$

In the real circuit, the total resistance is not only the externally added resistor, because the coil also has an internal resistance. Therefore, the effective total resistance is written as

$$R_{\text{total}} = R + R_{\text{sp}},$$

where R_{sp} is the internal resistance of the coil. This correction is important for comparing the measured damping and impedance with the theoretical model.

3 Experimental Analysis

3.1 Experimental Setup

The experiment was carried out as a HomeLab experiment using a Picoscope 2204A, an Arduino, a MOSFET switch, a breadboard, resistors, capacitors, a coil, and connecting cables. The Arduino was used to control the MOSFET, which periodically switched the discharge path of the capacitor. The voltage signal was measured with Channel A of the Picoscope across the capacitor. The Arduino supplied a voltage of 5 V.

For both the RC and RLC measurements, a fixed charging resistor of

$$R_{\text{ch}} = 22\,\text{k}\Omega$$

was used. This resistor charged the capacitor before each discharge process. When the MOSFET switched the discharge circuit on, the capacitor discharged through the selected circuit elements. Since the charging resistor remained connected during the discharge, it was included in the theoretical description of the RC circuit through the effective resistance

$$R_{\text{eff}} = R \parallel R_{\text{ch}}.$$

In the RC part of the experiment, a capacitor with capacitance

$$C_{\text{RC}} = 220\,\text{nF}$$

was discharged through five different resistors:

$$R = 220\,\Omega,\ 1\,\text{k}\Omega,\ 2\,\text{k}\Omega,\ 10\,\text{k}\Omega,\ 20\,\text{k}\Omega.$$

The voltage across the capacitor was measured as a function of time. The expected behaviour was an exponential decay with a resistance-dependent time constant.

In the RLC part of the experiment, a capacitor with capacitance

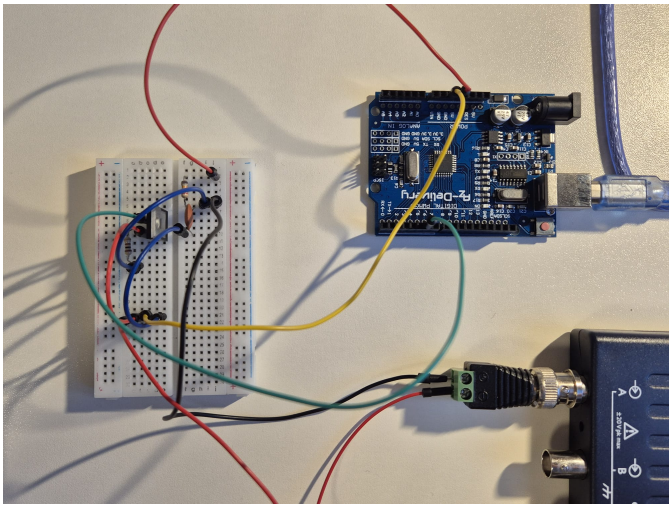
$$C_{\text{RLC}} = 1\,\text{nF}$$

was discharged through a series circuit consisting of a resistor, an inductor, and the capacitor. The five external resistances used were

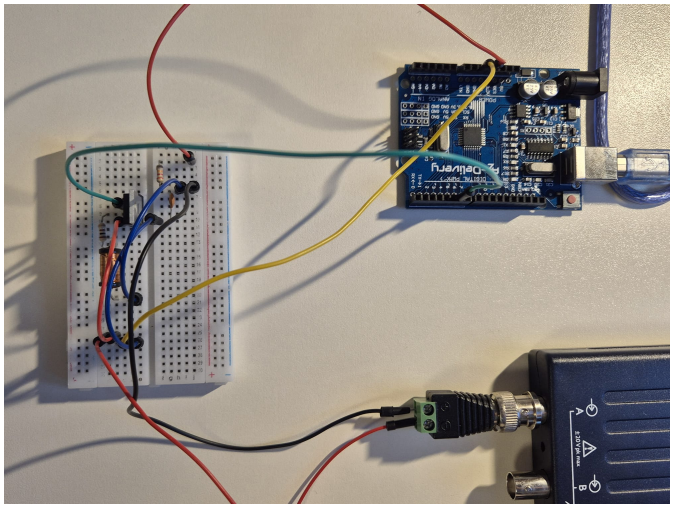
$$R = 10\,\Omega,\ 20\,\Omega,\ 100\,\Omega,\ 220\,\Omega,\ 320\,\Omega.$$

The inductance of the coil was not known directly and was therefore determined later from the measured damped oscillations. Since real coils are not ideal, the internal resistance of the coil was also considered in the interpretation of the results.

The Picoscope voltage range for Channel A was set to 10 V. The trigger was set to simple edge triggering in repeat mode, with a trigger level of approximately 20%. For most measurements, a trigger voltage of 2 V was used, while for some measurements a trigger voltage of 4 V was selected to obtain a stable curve. The time scale and sampling settings of the Picoscope were adjusted for each measurement so that the complete discharge curve could be recorded with sufficient resolution.

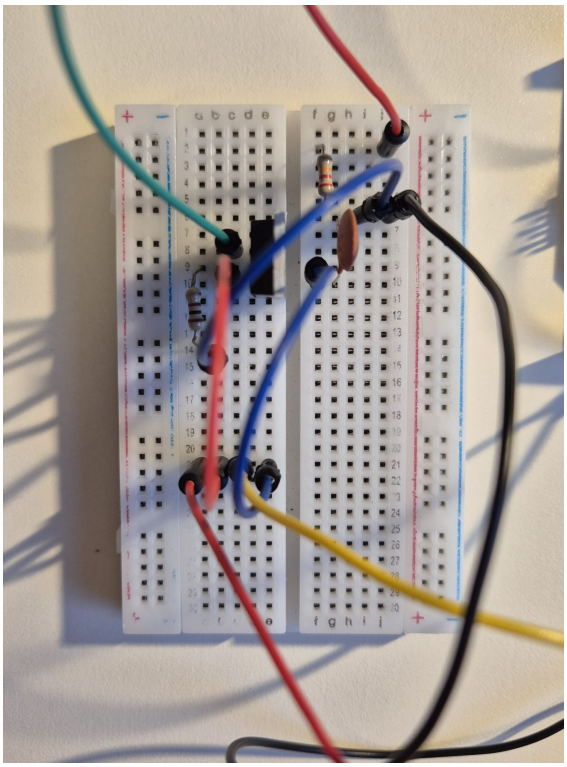


(a) RC circuit setup.

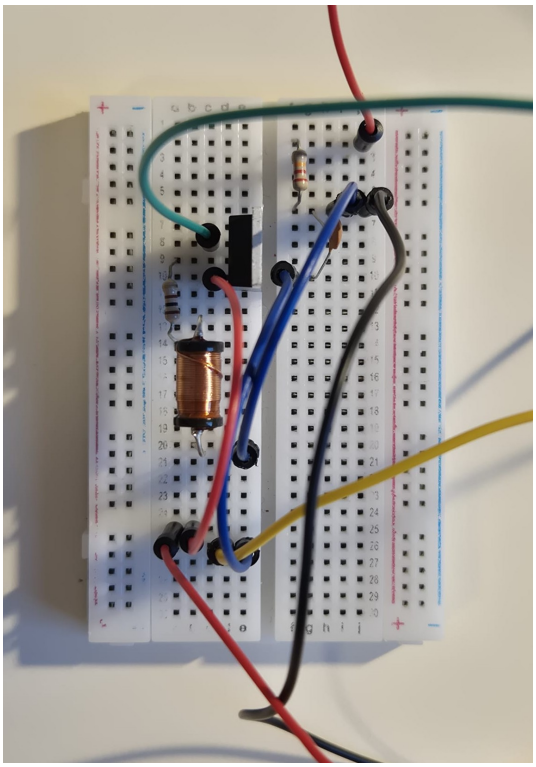


(b) RLC series circuit setup.

Figure 1: Experimental setups used for the RC and RLC discharge measurements.



(a) Close-up photograph of the RC circuit setup.



(b) Close-up photograph of the RLC series circuit setup.

Figure 2: Close-up photographs of the experimental setups used for the RC and RLC discharge measurements.

3.2 Data Acquisition and Measurement Procedure

For each measurement, the capacitor was first charged through the fixed 22 kΩ charging resistor using the 5 V Arduino output. The Arduino periodically changed the state of the MOSFET switch, which allowed the capacitor to discharge through the selected RC or RLC circuit. The switching event was used to trigger the Picoscope recording.

The measured signal was the capacitor voltage $U_C(t)$. For the RC circuit, this voltage was expected to decrease exponentially from its initial value toward a final offset voltage. The offset occurs because the charging resistor remained connected during the discharge, so the capacitor did not necessarily discharge completely to zero. Therefore, the RC data were later fitted using the function

$$U_C(t) = U_\infty + Ae^{-t/\tau}.$$

For the RLC circuit, the capacitor voltage showed a damped oscillation. This oscillatory behaviour is caused by the exchange of energy between the electric field of the capacitor and the magnetic field of the coil. The amplitude decreases with time because energy is dissipated in the external resistor and in the internal resistance of the coil. The RLC data were later fitted with the function

$$U_C(t) = U_{\text{off}} + Ae^{-\delta t}\sin(\omega_d t + \phi).$$

For the RC circuit, five separate measurements were performed with different discharge resistors:

$$220\,\Omega,\ 1\,\text{k}\Omega,\ 2\,\text{k}\Omega,\ 10\,\text{k}\Omega,\ 20\,\text{k}\Omega.$$

For the RLC circuit, five measurements were performed with the external resistors:

$$10\,\Omega,\ 20\,\Omega,\ 100\,\Omega,\ 220\,\Omega,\ 320\,\Omega.$$

For each resistance, the Picoscope time base was adjusted so that the relevant discharge process was clearly visible. Shorter time windows were used for faster discharges and longer time windows were used for slower discharges.

The recorded data were exported from the Picoscope software as CSV files. Each file contained a time column and the measured Channel A voltage. The data were then processed using Python. The time axis was shifted so that the beginning of the discharge corresponded approximately to $t = 0$. For the RC data, exponential fits were used to determine the time constants. For the RLC data, damped sinusoidal fits were used to determine the damping coefficient and damped angular frequency.

The frequency-dependent behaviour was obtained by applying a Fourier transform to the measured time-domain signals. Since the data were recorded at discrete time intervals, the FFT output was multiplied by the sampling

interval Δt in order to preserve the correct physical units of the Fourier transform. Only the positive frequency range was used for the impedance and phase analysis. The experimentally determined impedance magnitude and phase shift were then compared with the theoretical expressions for the RC and RLC circuits.

4 RC Circuit (Task 1)

4.1 Goal

The aim of the first task was to measure the discharge of a capacitor in an RC circuit for five different discharge resistances and to determine the corresponding time constants, impedance magnitudes, and phase shifts. The measured capacitor voltage was analysed in the time domain by fitting exponential decay functions. In addition, the discharge curves were transformed into the frequency domain in order to compare the experimentally determined impedance and phase shift with the theoretical behaviour of an RC circuit.

The capacitor used in this part of the experiment had the nominal capacitance

$$C = 220\,\mathrm{nF}.$$

The five discharge resistances were

$$R = 220\,\Omega,\ 1\,\mathrm{k}\Omega,\ 2\,\mathrm{k}\Omega,\ 10\,\mathrm{k}\Omega,\ 20\,\mathrm{k}\Omega.$$

In all measurements, the capacitor was charged through a fixed charging resistor

$$R_{\mathrm{ch}} = 22\,\mathrm{k}\Omega.$$

Since the charging resistor remained connected during the discharge process, the effective resistance relevant for the capacitor discharge was the parallel combination of the discharge resistor and the charging resistor:

$$R_{\mathrm{eff}} = R \parallel R_{\mathrm{ch}} = \frac{RR_{\mathrm{ch}}}{R + R_{\mathrm{ch}}}.$$

4.2 Theory and Relevant Equations

For a series RC circuit, the total complex impedance is

$$Z = Z_R + Z_C.$$

With

$$Z_R = R$$

and

$$Z_C = \frac{1}{i\omega C},$$

the total impedance is

$$Z = R + \frac{1}{i\omega C}.$$

Using $1/i = -i$, this can be written as

$$Z = R - \frac{i}{\omega C}.$$

The theoretical impedance magnitude is therefore

$$|Z| = \sqrt{R^2 + \frac{1}{\omega^2 C^2}},$$

and the phase shift is

$$\theta = -\arctan\left(\frac{1}{\omega RC}\right).$$

For the discharge analysis, the ideal capacitor voltage follows an exponential decay. Because the charging resistor remained connected, the theoretical time constant was calculated using R_{eff}:

$$\tau_{\mathrm{theory}} = R_{\mathrm{eff}} C.$$

The measured discharge curves were fitted with

$$U_C(t) = U_\infty + Ae^{-t/\tau},$$

where U_∞ is the final voltage offset, A is the amplitude, and τ is the fitted time constant. The offset term was included because the capacitor did not discharge fully to zero due to the remaining connection through the charging resistor.

For the frequency-domain analysis, the normalized discharge function can be written as

$$f_D(t) = e^{-t/\tau}.$$

Its Fourier transform is

$$\hat{f}_D(\omega) = \frac{\tau}{1 + i\omega\tau}.$$

For the RC series circuit, the impedance was determined using the relation

$$\frac{R}{Z_s} = i\omega \hat{f}_D(\omega),$$

so that

$$Z_s = \frac{R}{i\omega \hat{f}_D(\omega)}.$$

In the numerical FFT analysis, the Fourier transform was multiplied by the sampling interval Δt, since this factor is required to preserve the correct physical unit of the transform.

4.3 Measurement and Analysis Method

The capacitor voltage was measured across the capacitor using Channel A of the Picoscope. For each resistance, the capacitor was first charged by the 5 V Arduino output through the 22 kΩ charging resistor. The MOSFET then switched the discharge path on, allowing the capacitor to discharge through the selected resistor. The Picoscope was operated with a Channel A voltage range of 10 V. Simple edge triggering in repeat mode was used, with the trigger level set to approximately 20%. Depending on the measurement, a trigger voltage of approximately 2 V or 4 V was selected to obtain a stable discharge curve.

The measured data were exported as CSV files and analysed using Python. The discharge region was selected from each voltage curve and the time axis was shifted so that the beginning of the discharge corresponded to $t = 0$. Each curve was then fitted with the exponential function

$$U_C(t) = U_\infty + Ae^{-t/\tau}.$$

The fitted time constants were compared with the theoretical values calculated from

$$\tau_{\text{theory}} = R_{\text{eff}} C.$$

The impedance magnitude and phase shift were then determined from the Fourier transform of the normalized discharge signal and compared with the theoretical RC expressions.

4.4 Results

The raw RC discharge curves are shown in Fig. 3. The curves show the expected behaviour: the capacitor voltage decreases after the discharge path is switched on, and the discharge becomes slower as the resistance increases. For the larger resistances, the final voltage is clearly above zero, which is caused by the charging resistor remaining connected during the discharge.

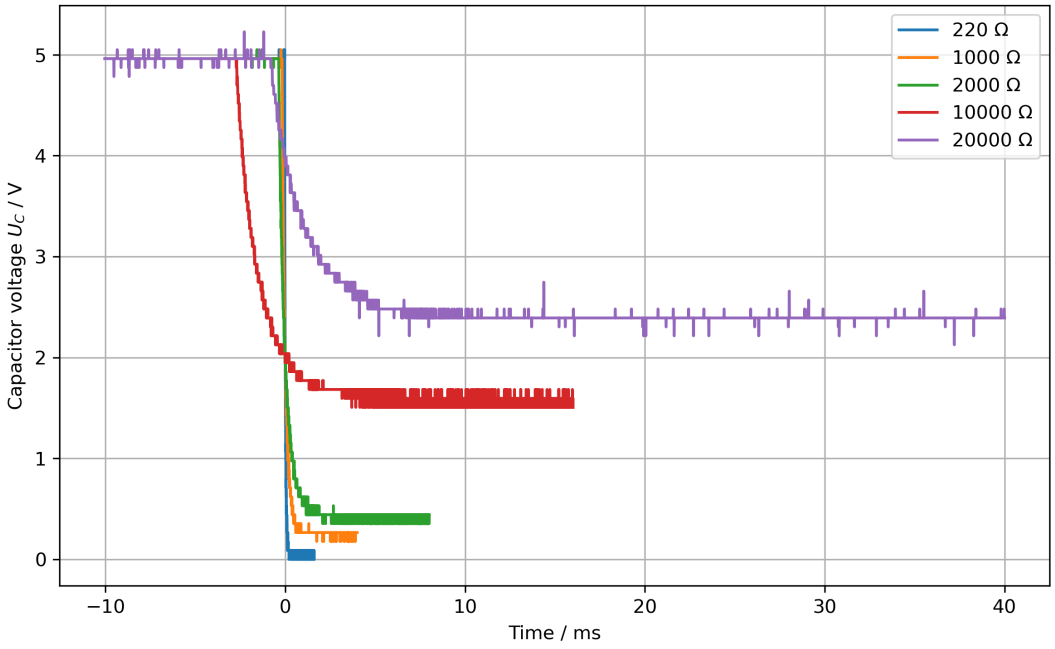


Figure 3: Raw RC discharge curves.

Figure 4 shows the selected discharge regions together with the fitted exponential curves. The fit function includes the final offset voltage U_∞, which is necessary because the capacitor voltage approaches a nonzero final value for the larger resistances.

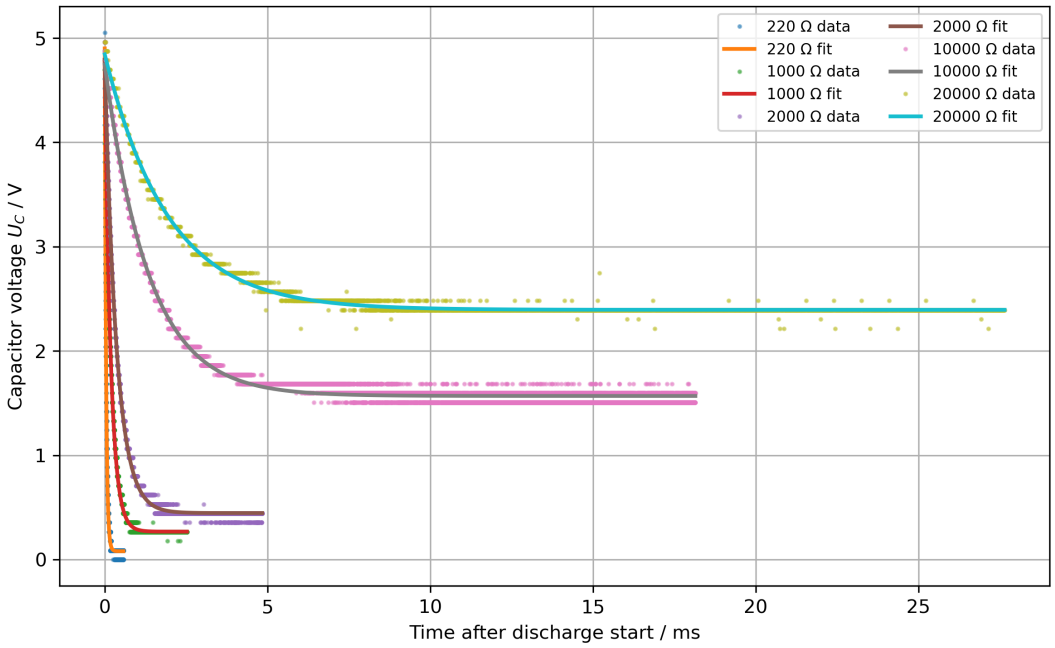


Figure 4: RC discharge curves with exponential fits using $U_C(t) = U_\infty + Ae^{-t/\tau}$.

The numerical fit results are listed in Table 1. The fitted time constants are compared with the theoretical values obtained from the effective resistance $R_{\text{eff}} = R \parallel 22\,\text{k}\Omega$.

Table 1: Experimental and theoretical time constants for the RC circuit.

R/Ω	R_{eff}/Ω	$u(R_{\text{eff}})/\Omega$	$\tau_{\text{fit}}/\text{ms}$	$u(\tau_{\text{fit}})/\text{ms}$	$\tau_{\text{theory}}/\text{ms}$	Deviation/%
220	217.82	6.23	0.0422	0.0001	0.0479	-11.92
1000	956.52	26.44	0.1848	0.0002	0.2104	-12.16
2000	1833.33	48.71	0.3623	0.0006	0.4033	-10.18
10000	6875.00	149.88	1.3494	0.0032	1.5125	-10.79
20000	10476.19	214.09	1.9392	0.0056	2.3048	-15.86

The comparison between the fitted and theoretical time constants is shown in Fig. 5. The fitted values follow the same trend as the theoretical prediction, confirming that the time constant increases with resistance. However, all measured values lie below the line of ideal agreement.

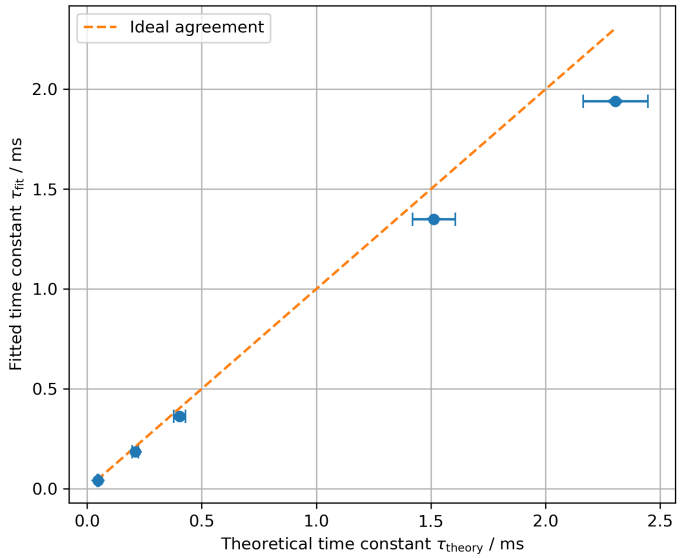


Figure 5: Comparison of fitted and theoretical RC time constants.

The theoretical impedance magnitude for the five effective resistances is shown in Fig. 6. At low frequencies, the capacitive reactance dominates and the impedance is large. At high frequencies, the capacitive reactance becomes small and the impedance approaches the effective resistance R_{eff}.

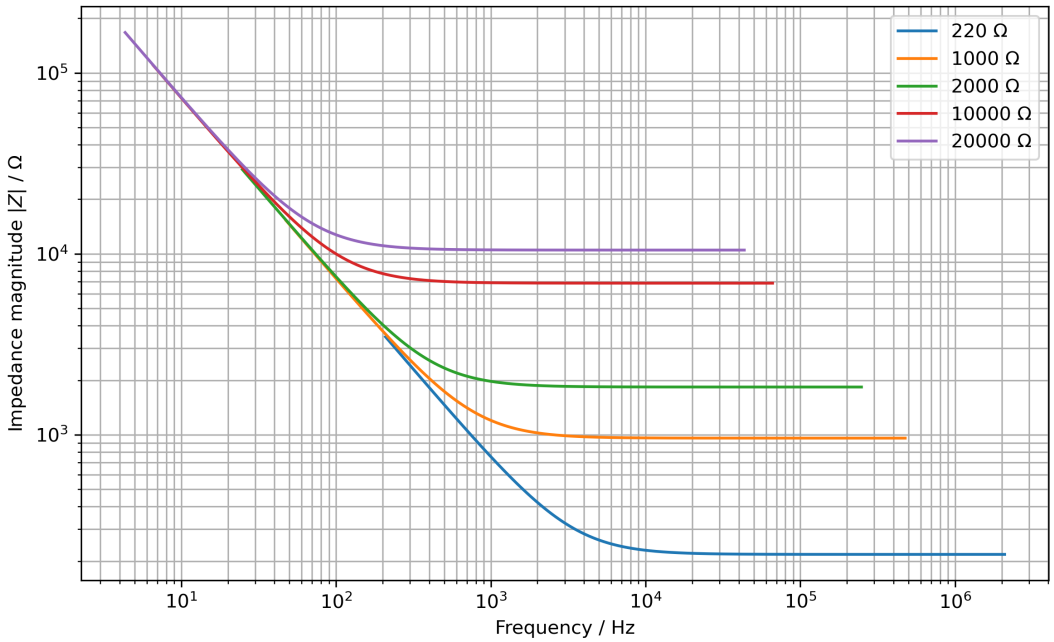


Figure 6: Theoretical impedance magnitude.

The corresponding theoretical phase shift is shown in Fig. 7. At low frequencies, the phase shift approaches $-90°$, which corresponds to dominant capacitive behaviour. At high frequencies, the phase shift approaches $0°$, meaning that the circuit behaves approximately like a pure resistor.

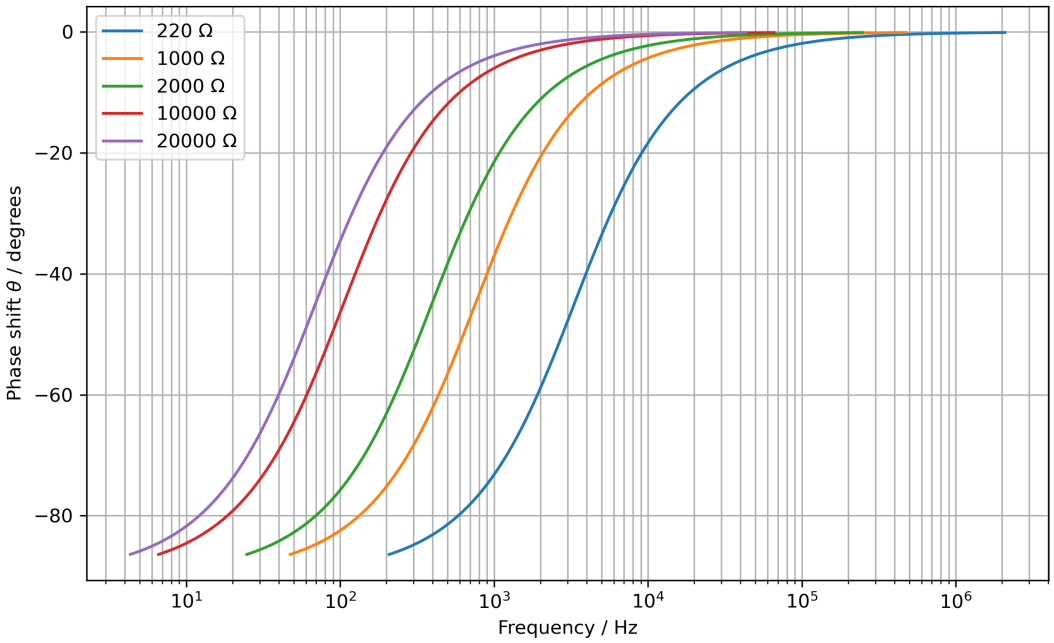


Figure 7: Theoretical phase shift.

The experimentally determined impedance magnitude from the Fourier-transform analysis is shown together with the theoretical curves in Fig. 8. The experimental values generally follow the expected trend and approach a resistance-dependent plateau at higher frequencies. Some scatter is visible, especially at higher frequencies, which is expected because the FFT method is sensitive to noise, offset correction, and the finite measurement window.

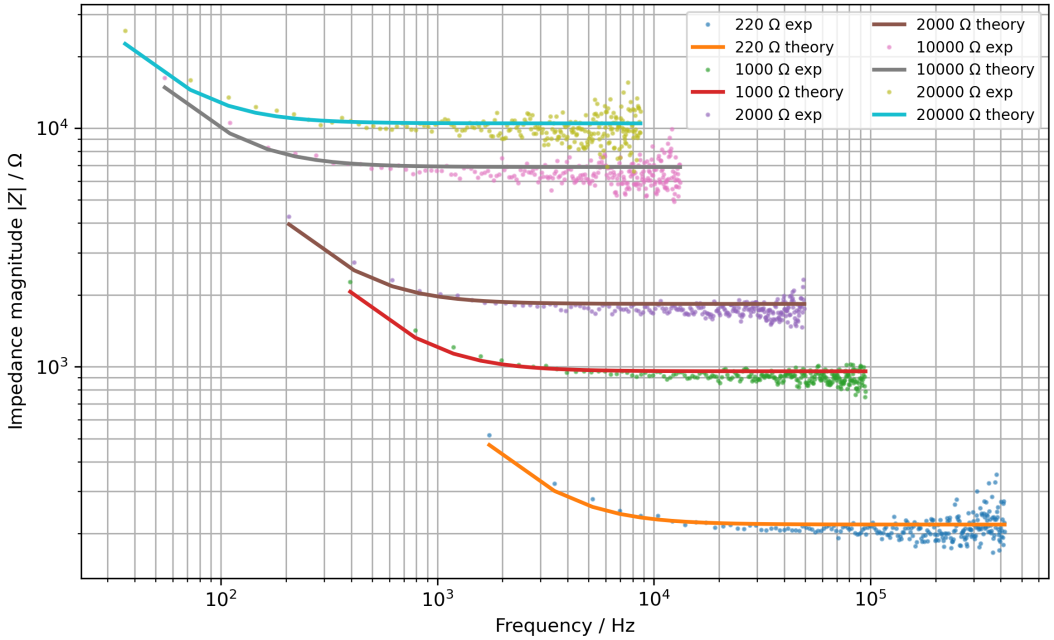


Figure 8: Experimental and theoretical RC impedance magnitude obtained from the Fourier-transform analysis.

The phase shift obtained from the Fourier-transform analysis is shown in Fig. 9. The experimental points show the same qualitative behaviour as the theoretical curves: the phase shift is negative and tends toward 0° at higher frequencies. The phase data show stronger scatter than the impedance magnitude because the phase is particularly sensitive to noise and offset errors in the Fourier transform.

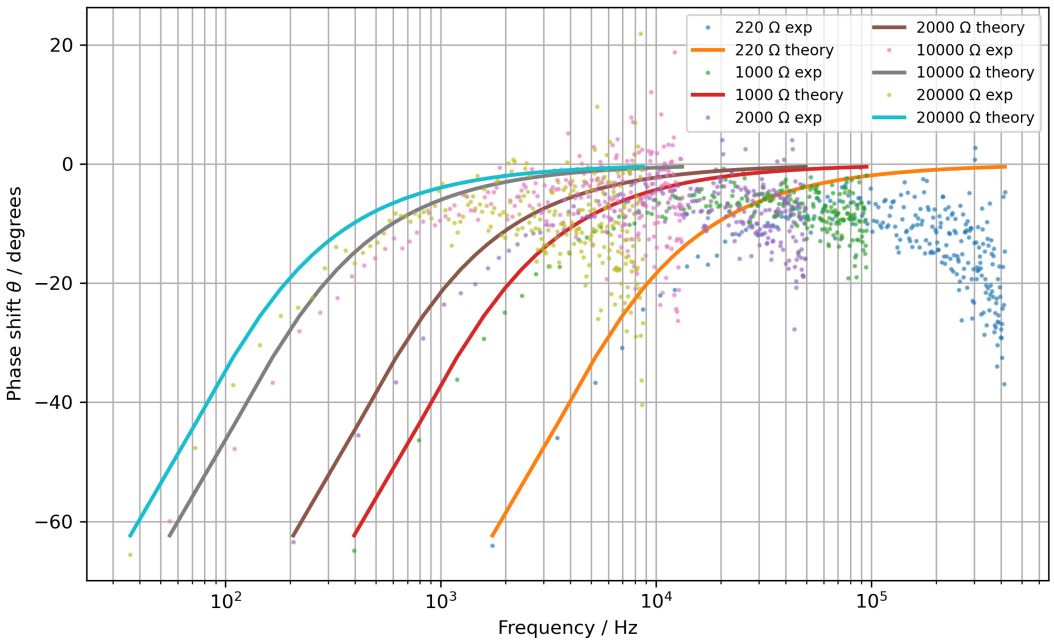


Figure 9: Experimental and theoretical RC phase shift obtained from the Fourier-transform analysis.

4.5 Uncertainty Analysis

The uncertainty of the resistance values was calculated from the given 5% resistor tolerance. Assuming a rectangular distribution, the standard uncertainty is

$$u(R) = \frac{0.05R}{\sqrt{3}}.$$

The same relative tolerance was assumed for the charging resistor. Since no tolerance was written on the capacitor, a 10% capacitance tolerance was assumed:

$$u(C) = \frac{0.10C}{\sqrt{3}}.$$

For the effective resistance

$$R_{\text{eff}} = \frac{RR_{\text{ch}}}{R + R_{\text{ch}}},$$

Gaussian error propagation was used:

$$u(R_{\text{eff}}) = \sqrt{\left(\frac{\partial R_{\text{eff}}}{\partial R} u(R)\right)^2 + \left(\frac{\partial R_{\text{eff}}}{\partial R_{\text{ch}}} u(R_{\text{ch}})\right)^2}.$$

The derivatives are

$$\frac{\partial R_{\text{eff}}}{\partial R} = \frac{R_{\text{ch}}^2}{(R + R_{\text{ch}})^2},$$

and

$$\frac{\partial R_{\text{eff}}}{\partial R_{\text{ch}}} = \frac{R^2}{(R + R_{\text{ch}})^2}.$$

The theoretical time constant uncertainty was calculated from

$$\tau_{\text{theory}} = R_{\text{eff}} C$$

using

$$u(\tau_{\text{theory}}) = \tau_{\text{theory}} \sqrt{\left(\frac{u(R_{\text{eff}})}{R_{\text{eff}}}\right)^2 + \left(\frac{u(C)}{C}\right)^2}.$$

The Picoscope voltage range was 10 V. Since the Picoscope has an 8-bit resolution, the voltage step was estimated as

$$\Delta V = \frac{10\,\text{V}}{2^8} = 0.039\,\text{V}.$$

The corresponding voltage uncertainty was estimated as

$$u(V) = \frac{\Delta V}{\sqrt{12}} \approx 0.011\,\text{V}.$$

This uncertainty affects the fitted decay curves, especially near the final voltage offset where the voltage changes slowly.

The uncertainty of the fitted time constant $u(\tau_{\text{fit}})$ was obtained from the covariance matrix of the exponential fit. These statistical fit uncertainties are small compared with the systematic uncertainty caused by component tolerances, especially the assumed capacitance tolerance.

4.6 Interpretation and Discussion

The RC measurements show the expected qualitative behaviour. Increasing the discharge resistance increases the time constant, so the capacitor discharges more slowly. The fitted exponential curves describe the selected discharge regions well, confirming that the capacitor voltage follows approximately exponential decay.

The fitted time constants are systematically lower than the theoretical values by approximately 10% to 16%. This systematic deviation suggests that the difference is not mainly caused by random noise. The most likely explanation is that the real capacitance was lower than the nominal value of 220 nF. This is plausible because no exact capacitor tolerance was available and a 10% tolerance had to be assumed. Additional contributions may come from resistor tolerances, the non-ideal resistance of the MOSFET path, contact resistances, and uncertainties in selecting the discharge start.

The nonzero final voltage in the discharge curves is also physically reasonable. Since the 22 kΩ charging resistor remained connected during the discharge, the capacitor did not discharge into a completely isolated resistor. Instead, the final voltage was determined by the circuit conditions after switching. This is why the fit function included the offset voltage U_∞.

The Fourier-transform analysis produced reasonable impedance and phase curves. The impedance magnitude decreases with increasing frequency and tends toward a resistance-dependent plateau, as expected for an RC circuit. The phase shift is negative and approaches zero at high frequencies. The experimental FFT-based phase values are noisier than the impedance values, which is expected because the phase is sensitive to noise, offset subtraction, and the finite time window used in the Fourier transform.

Overall, the RC part of the experiment confirms the expected exponential discharge behaviour and the qualitative frequency dependence of the impedance and phase shift. The remaining deviations from theory are consistent with component tolerances and non-ideal experimental conditions.

5 RLC Series Circuit (Task 2)

5.1 Goal

The aim of the second task was to investigate the discharge of a capacitor in an RLC series circuit for five different external resistances. In contrast to the RC circuit, the capacitor voltage in the RLC circuit shows a damped oscillation. This oscillation is caused by the periodic exchange of energy between the electric field of the capacitor and the magnetic field of the coil, while the amplitude decreases due to energy dissipation in the resistive parts of the circuit.

The capacitor used in this part had the nominal capacitance

$$C = 1\,\mathrm{nF}.$$

The five external resistances were

$$R = 10\,\Omega,\ 20\,\Omega,\ 100\,\Omega,\ 220\,\Omega,\ 320\,\Omega.$$

As in Task 1, the capacitor was charged through a fixed charging resistor

$$R_{\mathrm{ch}} = 22\,\mathrm{k}\Omega.$$

The inductance of the coil was not known directly. Therefore, it was determined from the fitted oscillation frequencies of the measured discharge curves. The internal resistance of the coil was also estimated from the damping behaviour.

5.2 Theory and Relevant Equations

For a series RLC circuit, Kirchhoff's loop rule gives

$$U_L + U_R + U_C = 0$$

after the voltage supply has been switched off. Using

$$U_L = L\frac{dI}{dt}, \qquad U_R = RI, \qquad U_C = \frac{Q}{C},$$

and

$$I = \frac{dQ}{dt},$$

one obtains the differential equation

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = 0.$$

Dividing by L gives

$$\frac{d^2Q}{dt^2} + \frac{R}{L}\frac{dQ}{dt} + \frac{1}{LC}Q = 0.$$

The natural angular frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}},$$

and the damping coefficient is

$$\delta = \frac{R_{\mathrm{total}}}{2L}.$$

Here,

$$R_{\mathrm{total}} = R + R_{\mathrm{sp}}$$

is the total resistance of the oscillating circuit, including the external resistor R and the internal resistance of the coil R_{sp}.

For the underdamped case, the solution is a damped oscillation. Since the measured capacitor voltage is proportional to charge,

$$U_C(t) = \frac{Q(t)}{C},$$

the measured voltage was fitted using

$$U_C(t) = U_{\mathrm{off}} + Ae^{-\delta t}\cos(\omega_d t + \phi),$$

where U_{off} is the voltage offset, A is the amplitude, δ is the damping coefficient, ω_d is the damped angular frequency, and ϕ is a phase constant.

The damped angular frequency is related to the natural angular frequency by

$$\omega_d = \sqrt{\omega_0^2 - \delta^2}.$$

Therefore,

$$\omega_0 = \sqrt{\omega_d^2 + \delta^2}.$$

The inductance can then be calculated from

$$L = \frac{1}{C\omega_0^2}.$$

The complex impedance of the RLC series circuit is

$$Z = Z_R + Z_L + Z_C.$$

Using

$$Z_R = R_{\text{total}}, \qquad Z_L = i\omega L, \qquad Z_C = \frac{1}{i\omega C},$$

the total impedance becomes

$$Z = R_{\text{total}} + i\left(\omega L - \frac{1}{\omega C}\right).$$

The impedance magnitude is therefore

$$|Z| = \sqrt{R_{\text{total}}^2 + \left(\omega L - \frac{1}{\omega C}\right)^2},$$

and the phase shift is

$$\theta = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R_{\text{total}}}\right).$$

At resonance,

$$\omega L = \frac{1}{\omega C},$$

so the inductive and capacitive reactances cancel. The resonance frequency is therefore

$$f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

At this frequency, the impedance reaches a minimum and is approximately equal to R_{total}, while the phase shift crosses $0°$.

5.3 Measurement and Analysis Method

The capacitor voltage $U_C(t)$ was measured across the capacitor using Channel A of the Picoscope. The Arduino supplied 5 V and controlled the MOSFET switch, which started the discharge through the RLC series circuit. The Picoscope was operated with a Channel A voltage range of 10 V. The trigger was set to simple edge triggering in repeat mode.

For each of the five external resistances, the measured voltage signal was exported as a CSV file and analysed using Python. The start of the oscillation was determined from the largest change in the measured voltage signal, and the time axis was shifted so that the beginning of the oscillation corresponded to $t = 0$. The selected oscillation region was then fitted using

$$U_C(t) = U_{\text{off}} + Ae^{-\delta t}\cos(\omega_d t + \phi).$$

From the fitted values of δ and ω_d, the natural angular frequency was calculated using

$$\omega_0 = \sqrt{\omega_d^2 + \delta^2}.$$

The coil inductance was then obtained from

$$L = \frac{1}{C\omega_0^2}.$$

The final inductance was taken as the mean value of the five individual estimates. The internal resistance of the coil was estimated using

$$R_{\text{sp}} = 2L\delta - R.$$

The impedance and phase shift were calculated using the fitted inductance and the estimated coil resistance. For the FFT-based comparison, the measured signal was first converted into a current-like signal. Since

$$Q = CU_C$$

and

$$I = \frac{dQ}{dt} = C\frac{dU_C}{dt},$$

the current was estimated by differentiating the fitted capacitor voltage. This method was used to obtain a qualitative frequency-domain comparison with the theoretical impedance and phase curves.

5.4 Results

The raw RLC discharge curves are shown in Fig. 10. All curves show damped oscillations. The oscillation amplitude decreases faster for larger external resistances, as expected from the damping term.

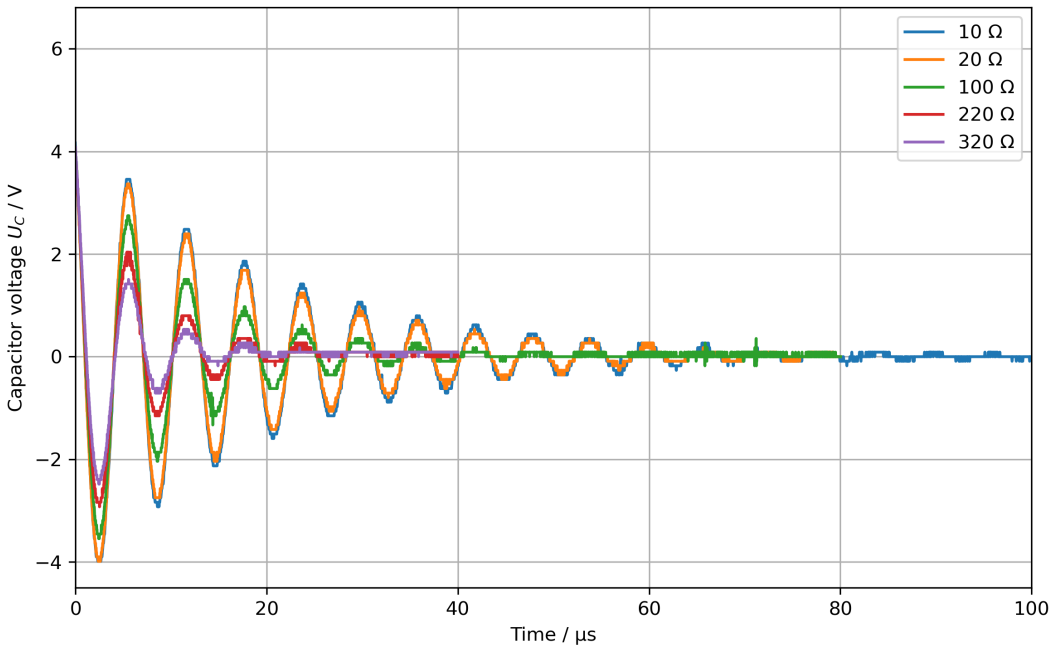


Figure 10: Raw RLC discharge curves.

The fitted damped oscillations are shown in Fig. 11. The fit function describes the measured oscillations well for all five resistances. This shows that the underdamped RLC model is appropriate for the measured discharge signals.

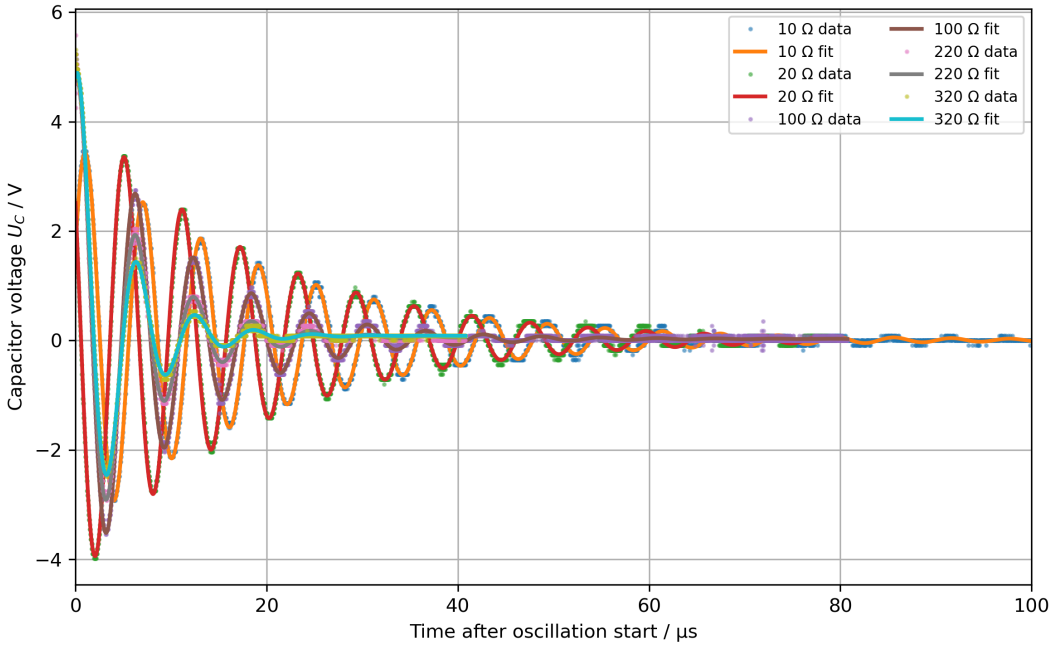


Figure 11: RLC discharge curves with damped cosine fits using $U_C(t) = U_{\text{off}} + Ae^{-\delta t}\cos(\omega_d t + \phi)$.

The fitted parameters and the resulting inductance values are listed in Table 2. The fitted damped frequencies are close to each other for all five measurements, while the damping coefficient increases with increasing resistance.

Table 2: Fitted parameters and calculated inductance values for the RLC circuit.

R/Ω	δ/s^{-1}	$u(\delta)/\mathrm{s}^{-1}$	f_d/kHz	f_0/kHz	L/mH	R_{sp}/Ω
10	5.05×10^4	9.69×10^1	165.51	165.70	0.923	83.2
20	5.58×10^4	9.42×10^1	164.85	165.09	0.929	82.9
100	9.53×10^4	1.55×10^2	164.62	165.32	0.927	75.9
220	1.53×10^5	2.13×10^2	164.26	166.06	0.919	63.3
320	2.07×10^5	2.59×10^2	162.69	165.98	0.919	61.1

The individual inductance values are shown in Fig. 12. The values are consistent with each other, giving the mean result

$$L = (0.923 \pm 0.024)\,\mathrm{mH}.$$

This value was used for the theoretical impedance and phase calculations.

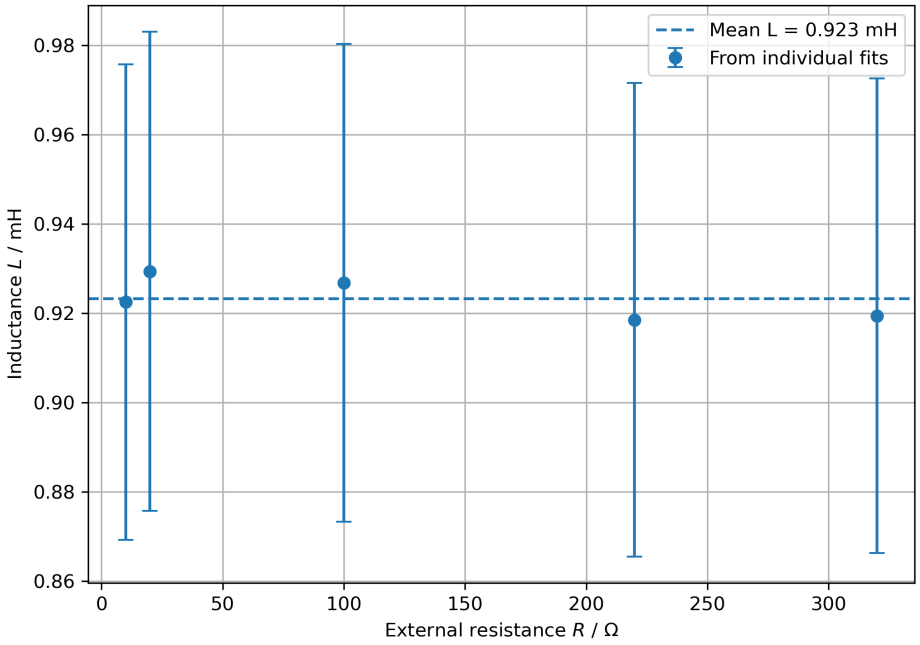


Figure 12: Inductance values obtained from the individual RLC fits.

The dependence of the damping coefficient on the external resistance is shown in Fig. 13. The approximately linear trend agrees with

$$\delta = \frac{R + R_{\mathrm{sp}}}{2L}.$$

From the individual damping values, the internal coil resistance was estimated as

$$R_{\mathrm{sp}} = (73 \pm 10)\,\Omega.$$

The linear fit of δ versus R gave a somewhat larger estimate of approximately

$$R_{\mathrm{sp}} \approx 91\,\Omega.$$

This difference is treated as a systematic uncertainty caused by the simplified model of the real coil and the sensitivity of the damping fit.

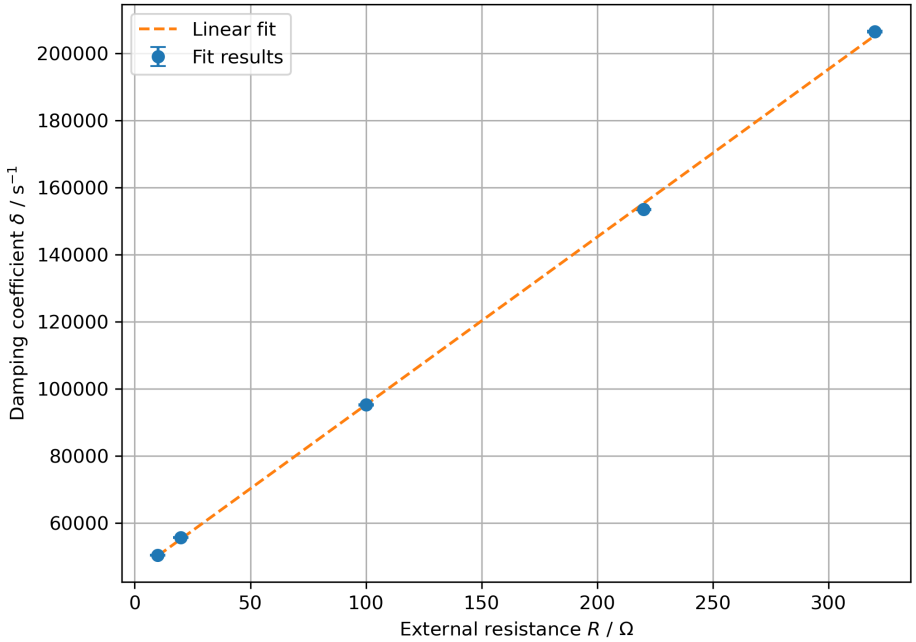


Figure 13: Damping coefficient as a function of external resistance.

The theoretical impedance magnitude of the RLC circuit is shown in Fig. 14. A clear impedance minimum occurs near the resonance frequency. The minimum impedance increases with increasing resistance because the total resistance of the circuit is larger.

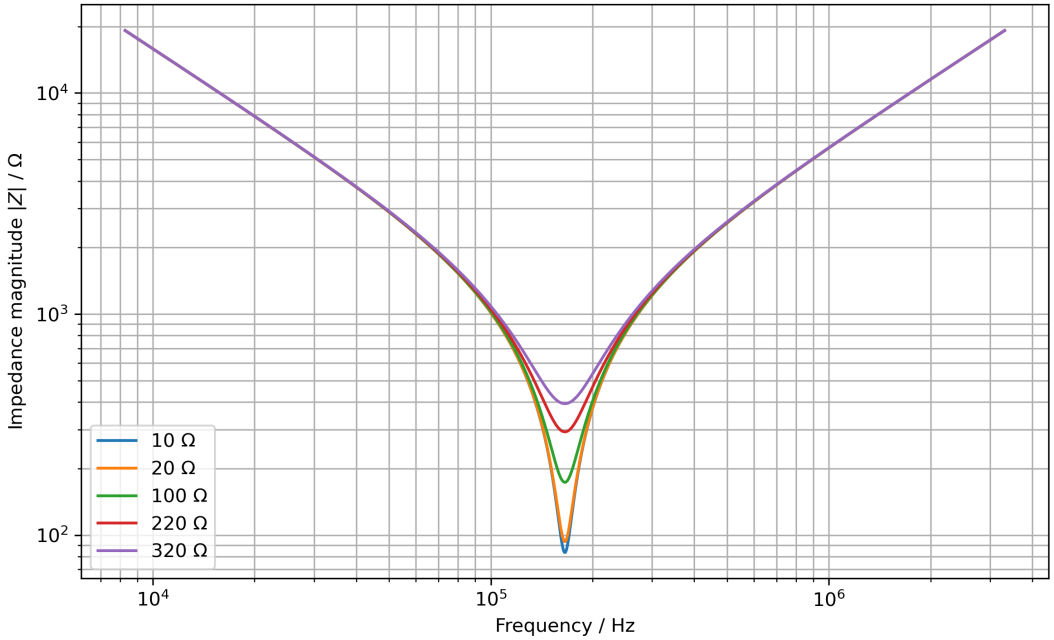


Figure 14: Theoretical impedance magnitude.

The theoretical phase shift is shown in Fig. 15. At frequencies below resonance, the circuit behaves mainly capacitively and the phase shift is negative. At frequencies above resonance, the circuit behaves mainly inductively and the phase shift becomes positive. Near resonance, the phase shift crosses approximately 0°.

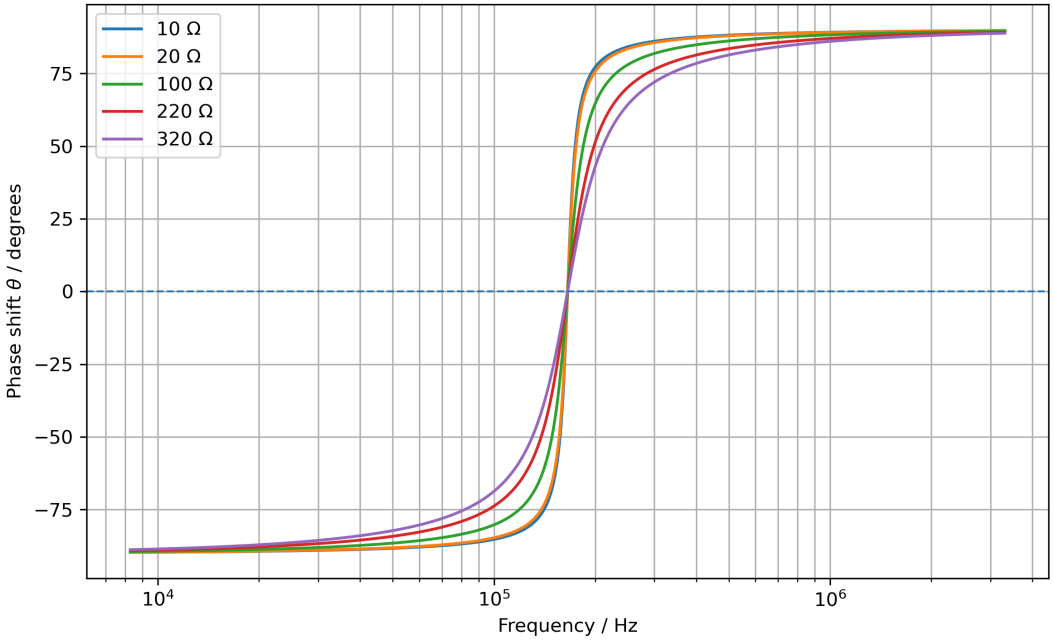


Figure 15: Theoretical phase shift.

The FFT-based impedance magnitude is compared with the theoretical curves in Fig. 16. The experimental curves show a minimum near the expected resonance region and follow the general shape of the theoretical impedance. Deviations are visible because the measured signal was the capacitor voltage and the current had to be estimated by numerical differentiation.

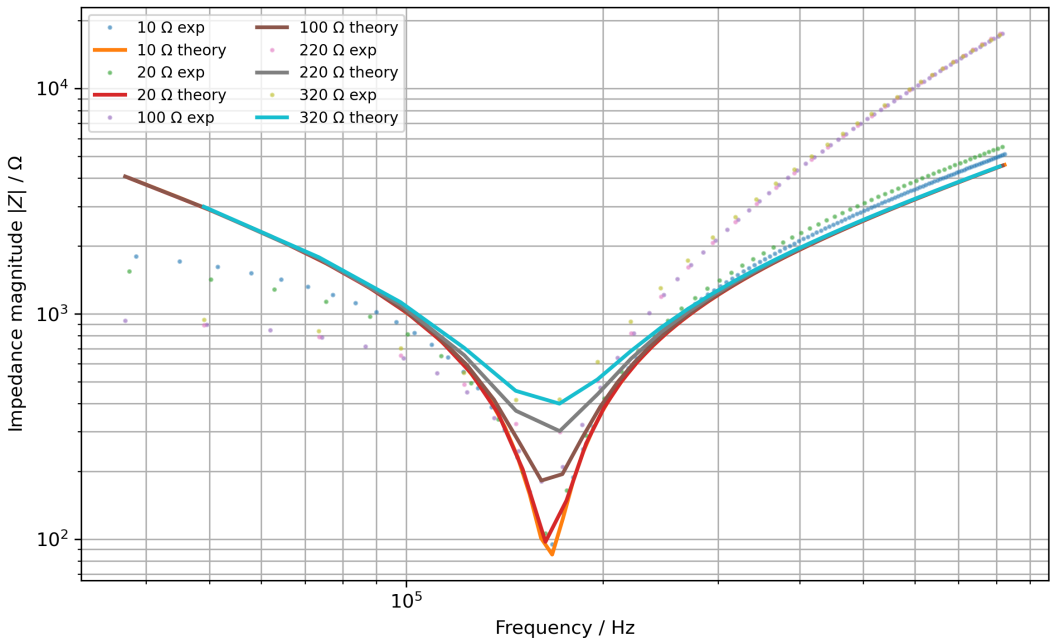


Figure 16: Experimental and theoretical RLC impedance magnitude obtained from the Fourier-transform analysis.

The FFT-based phase shift is shown in Fig. 17. The theoretical phase shift changes from negative to positive values around resonance. The experimental phase data show the same general transition, but with noticeable scatter and outliers. This is expected because the phase is sensitive to noise, the finite time window, phase wrapping, and the numerical differentiation used to estimate the current signal.

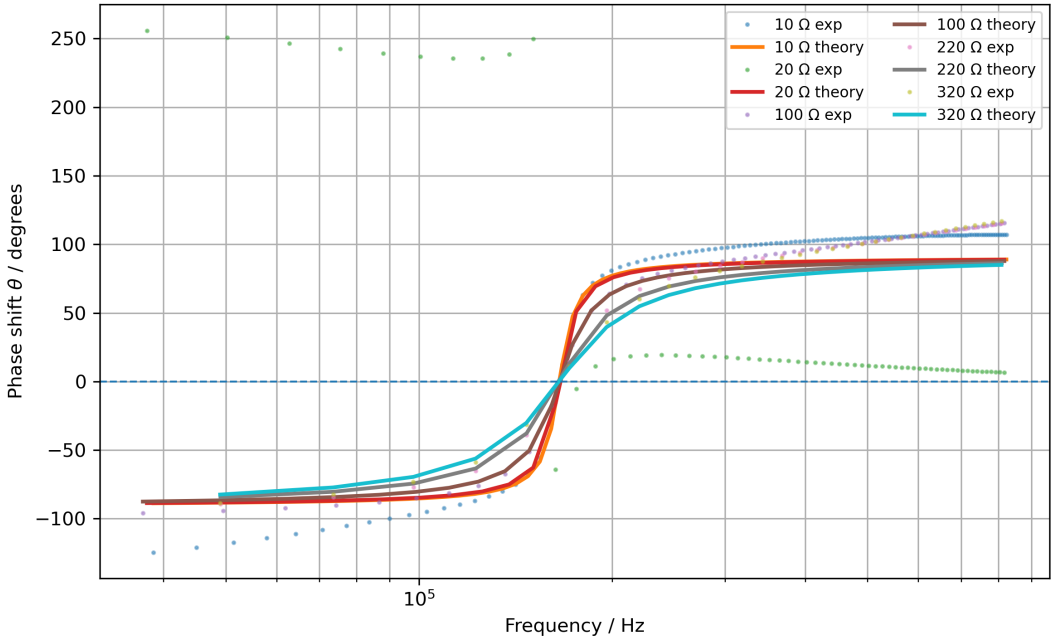


Figure 17: Experimental and theoretical RLC phase shift obtained from the Fourier-transform analysis.

5.5 Uncertainty Analysis

The resistance uncertainty was calculated from the 5% tolerance of the resistors. Assuming a rectangular distribution,

$$u(R) = \frac{0.05R}{\sqrt{3}}.$$

Since no tolerance was written on the capacitor, a 10% uncertainty was assumed:

$$u(C) = \frac{0.10C}{\sqrt{3}}.$$

The damping coefficient δ and damped angular frequency ω_d were obtained from the damped cosine fit. Their statistical uncertainties were taken from the covariance matrix of the fit.

The natural angular frequency was calculated from

$$\omega_0 = \sqrt{\omega_d^2 + \delta^2}.$$

The inductance was then calculated using

$$L = \frac{1}{C\omega_0^2}.$$

The uncertainty of L was obtained by error propagation:

$$u(L) = \sqrt{\left(\frac{\partial L}{\partial C}u(C)\right)^2 + \left(\frac{\partial L}{\partial \omega_d}u(\omega_d)\right)^2 + \left(\frac{\partial L}{\partial \delta}u(\delta)\right)^2}.$$

The partial derivatives are

$$\frac{\partial L}{\partial C} = -\frac{1}{C^2\omega_0^2},$$

$$\frac{\partial L}{\partial \omega_d} = -\frac{2\omega_d}{C\omega_0^4},$$

and

$$\frac{\partial L}{\partial \delta} = -\frac{2\delta}{C\omega_0^4}.$$

The internal resistance of the coil was estimated from

$$R_{\mathrm{sp}} = 2L\delta - R.$$

Its uncertainty was estimated using

$$u(R_{\mathrm{sp}}) = \sqrt{\left(2\delta u(L)\right)^2 + \left(2Lu(\delta)\right)^2 + u(R)^2}.$$

The Picoscope voltage uncertainty was estimated from the 10 V range and the 8-bit resolution:

$$\Delta V = \frac{10\,\mathrm{V}}{2^8} = 0.039\,\mathrm{V},$$

and therefore

$$u(V) = \frac{\Delta V}{\sqrt{12}} \approx 0.011\,\mathrm{V}.$$

This contributes to the scatter in the fitted oscillation and especially affects the later part of the signal, where the amplitude is small.

5.6 Interpretation and Discussion

The measured RLC curves clearly show damped oscillations, confirming that the circuit was underdamped for all five external resistances. As expected, the damping increases with resistance. This agrees with the theoretical relation

$$\delta = \frac{R_{\mathrm{total}}}{2L}.$$

The fitted damped frequencies were all close to 165 kHz, and the calculated inductance values were consistent across the five measurements. This gives confidence in the fitted value

$$L = (0.923 \pm 0.024)\,\mathrm{mH}.$$

The difference between the estimated internal coil resistance from individual damping values and from the linear fit indicates that the real coil cannot be described perfectly by a constant ohmic resistance. Frequency-dependent coil resistance, parasitic capacitance, contact resistance, and uncertainties in the fitted damping coefficient may all contribute to this difference.

The theoretical impedance curves show the expected resonance minimum. At low frequencies, the circuit behaves mainly capacitively, while at high frequencies it behaves mainly inductively. The phase shift crosses approximately zero at the resonance frequency. The FFT-based impedance and phase results reproduce the general behaviour, but with visible scatter. This is expected because the measured quantity was the capacitor voltage, not the source voltage and current directly. Therefore, the current had to be estimated using

$$I = C\frac{dU_C}{dt},$$

which amplifies noise and makes the phase analysis less stable.

Overall, Task 2 confirms the theoretical behaviour of a damped RLC series circuit. The time-domain fits provide a reliable estimate of the coil inductance and show that the damping increases with resistance. The frequency-domain analysis also shows the expected resonance behaviour, although the experimental phase and impedance values are affected by noise and the indirect determination of the current.

6 Discussion

The aim of the experiment was to use time-resolved discharge measurements to determine the impedance magnitude and phase shift of RC and RLC circuits. Overall, the results show that the measured time-domain signals contain the expected information about the frequency-dependent behaviour of the circuits. The RC circuit showed exponential capacitor discharge, while the RLC circuit showed damped oscillations caused by the exchange of energy between the capacitor and the coil.

For the RC circuit, the measured voltage curves followed the expected exponential decay. The discharge became slower as the resistance increased, which agrees with the theoretical relation

$$\tau = R_{\mathrm{eff}}C.$$

It was important to use the effective resistance

$$R_{\mathrm{eff}} = R \parallel R_{\mathrm{ch}}$$

instead of only the selected discharge resistor, because the 22 kΩ charging resistor remained connected during the discharge. This also explains the nonzero final voltage offset in the measured curves. The fitted time constants were systematically smaller than the theoretical values by about 10% to 16%. Since all deviations have the same sign, the difference is likely caused mainly by a systematic effect rather than random noise. A likely reason is that the real capacitance was smaller than the nominal value of 220 nF. This is plausible because no exact capacitor tolerance was available and a 10% tolerance had to be assumed. Further contributions may come from resistor tolerances, contact

resistances, the MOSFET switching path, and uncertainties in the selection of the discharge region.

The Fourier-transform analysis of the RC circuit reproduced the expected qualitative behaviour. The impedance magnitude decreased with increasing frequency and approached a resistance-dependent plateau at high frequencies. The phase shift was negative and approached $0°$ at high frequencies, as expected for a circuit that becomes increasingly resistive when the capacitive reactance decreases. The FFT-based results showed more scatter than the fitted time constants. This is expected because the Fourier transform is sensitive to offset correction, the finite time window, voltage noise, and the sampling resolution of the Picoscope.

For the RLC circuit, all five measurements showed clear damped oscillations. This confirms that the circuit was underdamped for all selected resistances. Increasing the external resistance increased the damping of the oscillation, which agrees with

$$\delta = \frac{R_{\text{total}}}{2L}.$$

The damped frequencies were very similar for all measurements, while the damping coefficient increased with resistance. From the fitted oscillation frequencies, the inductance was determined as

$$L = (0.923 \pm 0.024)\,\text{mH}.$$

The consistency of the individual inductance values indicates that the damped oscillation model described the measured data well.

The damping analysis also showed that the coil had a significant internal resistance. From the individual damping values, the internal resistance was estimated as

$$R_{\text{sp}} = (73 \pm 10)\,\Omega.$$

A linear fit of the damping coefficient as a function of external resistance gave a somewhat larger value of about

$$R_{\text{sp}} \approx 91\,\Omega.$$

The difference between these estimates shows that the real coil is not perfectly described by an ideal inductance with a constant series resistance. Possible reasons are frequency-dependent coil resistance, parasitic capacitance, contact resistance, and uncertainties in the fitted damping coefficients. This agrees with the expectation that real coils are non-ideal components.

The theoretical RLC impedance curves showed the expected resonance behaviour. The impedance had a minimum close to the resonance frequency, where the inductive and capacitive reactances cancel. At frequencies below resonance, the circuit behaved mainly capacitively and the phase shift was negative. Above resonance, the circuit behaved mainly inductively and the phase shift became positive. The FFT-based RLC impedance analysis showed the same general resonance behaviour, although the experimental points were noisier than the theoretical curves. The phase analysis was especially sensitive to noise and showed visible scatter and outliers. This is mainly because the Picoscope measured the capacitor voltage rather than the current directly. Therefore, the current had to be estimated using

$$I = C\frac{dU_C}{dt},$$

and numerical differentiation amplifies noise.

The main experimental limitations were the finite 8-bit voltage resolution of the Picoscope, component tolerances, the unknown exact capacitance values, the internal resistance and non-ideal behaviour of the coil, and the finite measurement windows used for the Fourier transforms. The trigger position and the exact choice of fitting interval also affected the results. Despite these limitations, the measured data were sufficient to confirm the expected behaviour of both circuits.

Overall, the experiment demonstrated that time-domain measurements can be used to obtain frequency-domain information about AC circuits. The RC circuit results confirmed exponential relaxation and capacitive phase behaviour, while the RLC circuit results showed damped oscillations, resonance, and the effect of increasing resistance on damping. The agreement with theory was good qualitatively and reasonable quantitatively, with the remaining deviations explained by component tolerances and non-ideal experimental conditions.

7 Conclusion

In this experiment, the time-dependent discharge behaviour of RC and RLC circuits was investigated using a Picoscope and a MOSFET switching circuit. From the measured capacitor voltage curves, characteristic circuit parameters were

determined and compared with theoretical predictions.

For the RC circuit, the capacitor discharge followed the expected exponential behaviour. The fitted time constants increased with resistance and were compared with

$$\tau_{\text{theory}} = R_{\text{eff}} C,$$

where the effective resistance included the parallel contribution of the 22 kΩ charging resistor. The fitted values were systematically about 10% to 16% smaller than theory, which can be explained mainly by capacitor tolerance and other non-ideal circuit effects. The Fourier-transform analysis also showed the expected decrease of impedance with frequency and the phase shift approaching 0° at high frequencies.

For the RLC circuit, damped oscillations were observed for all five external resistances. The damping coefficient increased with resistance, as predicted by

$$\delta = \frac{R_{\text{total}}}{2L}.$$

From the fitted oscillation frequencies, the coil inductance was determined as

$$L = (0.923 \pm 0.024)\,\text{mH}.$$

The internal resistance of the coil was estimated to be of the order of 70–90 Ω, confirming that the coil was not an ideal component. The impedance and phase results showed the expected resonance behaviour, with an impedance minimum and a phase transition near the resonance frequency.

Overall, the experiment confirmed that time-domain discharge measurements can be used to determine frequency-domain properties such as impedance and phase shift. The results agreed well with the theoretical models qualitatively and reasonably well quantitatively. The remaining deviations can be attributed mainly to component tolerances, non-ideal coil behaviour, finite Picoscope resolution, and limitations of the FFT analysis.

8 References

M. Ziese, E13e Time-resolved spectroscopy and impedances, University of Leipzig, June 2026.